[1]:
```python
import pandas as pd
import numpy as np

# World Happiness Report 2023 — representative data
# Source: https://www.kaggle.com/datasets/ajaypalsinghlo/
 world-happiness-report-2023

df = pd.read_csv('world_happiness_2023 (1).csv')
df.columns = ['Country','Region','Happiness_Score','GDP','Social_Support',
              'Life_Expectancy','Freedom','Generosity','Corruption']


print(f"Dataset: {len(df)} countries, {len(df.columns)} columns")
print(df.head())
```

```
Dataset: 63 countries, 9 columns
      Country                    Region Happiness_Score    GDP \
0     Finland             Western Europe          7.804
                                                  10.775
1     Denmark             Western Europe          7.586
                                                  10.933
2     Iceland             Western Europe          7.525
                                                  10.878
3    Israel Middle East and North Africa          7.473
                                                  10.527
4 Netherlands             Western Europe          7.464
                                                  11.015
   Social_Support Life_Expectancy Freedom Generosity Corruption
0           0.954           71.9   0.949      0.142      0.179
1           0.954           72.7   0.931      0.168      0.234
2           0.983           72.5   0.961      0.260      0.150
3           0.916           72.4   0.903      0.149      0.826
4           0.939           72.4   0.879      0.240      0.296
```

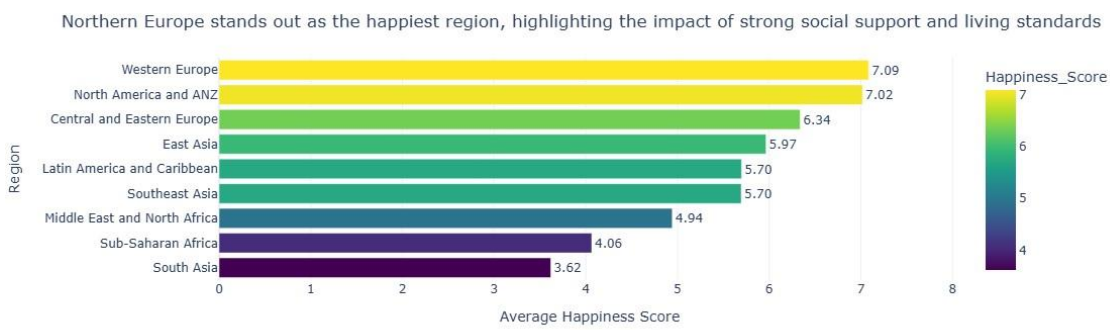
[2]:
```python
import plotly.express as px
import plotly.graph_objects as go

# Explore the dataset before you start
print("Regions in dataset:")
print(df['Region'].value_counts())
print("\nScore range:", df['Happiness_Score'].min(), "-", df['Happiness_Score'].
 max())
print("\nBottom 10 countries:")
print(df.nsmallest(10,
```

```
 'Happiness_Score')[['Country','Region','Happiness_Score']])
```

```
Matplotlib is building the font cache; this may take a moment.
Regions in dataset:
Western Europe              15
Latin America and Caribbean 13
Central and Eastern Europe   7
Sub-Saharan Africa           7
Middle East and North        6
Africa
North America and ANZ        4
Southeast Asia               4
South Asia                   4
East Asia                    3
Name: Region, dtype: int64

Score range: 1.859 - 7.804

Bottom 10 countries:
       Country                  Region Happiness_Score
60 Afghanistan                South Asia         1.859
61     Lebanon Middle East and North Africa 2.392
62     Zimbabwe  Sub-Saharan Africa    2.995
52    Ethiopia  Sub-Saharan Africa    3.564
53    Tanzania  Sub-Saharan Africa    3.698
48  Bangladesh                South Asia         3.892
47       India                South Asia         4.036
50 Kenya Sub-Saharan Africa 4.112 54 Uganda Sub-Saharan
Africa 4.333
55       Egypt  Middle East and North Africa            4.431
```

[4]:
```
# Step 1: Compute average happiness score by region
region_avg = (df.groupby('Region')['Happiness_Score']
              .mean()
              .reset_index()
              .sort_values('Happiness_Score'))
fig = px.bar(
    region_avg,
    x='Happiness_Score',
    y='Region',
    orientation='h',
    text='Happiness_Score',  # show values on bars
    color='Happiness_Score',  # add color gradient
    color_continuous_scale='Viridis'  # design choice beyond default
)
```

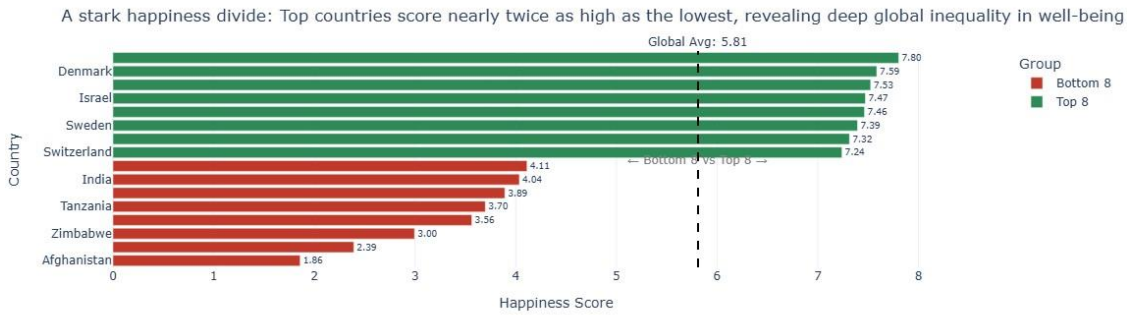
```
# Ensure descending order (highest at top)
fig.update_layout(
    yaxis=dict(categoryorder='total ascending'),  # because horizontal
    xaxis=dict(title='Average Happiness Score', range=[0,
 region_avg['Happiness_Score'].max() + 1]),
    title='Northern Europe stands out as the happiest region, highlighting the
 impact of strong social support and living standards ',
    template='plotly_white'
)

# Improve labels and readability
fig.update_traces(
    texttemplate='%{text:.2f}',
    textposition='outside'
)

fig.show()
```



[8]:
```
#Step 1: Get top and bottom countries
top8 = df.nlargest(8, 'Happiness_Score').copy()
top8['Group'] = 'Top 8'
bottom8 = df.nsmallest(8, 'Happiness_Score').copy()
bottom8['Group'] = 'Bottom 8'

combined = pd.concat([bottom8, top8]).sort_values('Happiness_Score')
global_avg = df['Happiness_Score'].mean()
print(f"Global average: {global_avg:.2f}")
fig = px.bar(
    combined,
    x='Happiness_Score',
    y='Country',
    orientation='h',
    color='Group',
    text='Happiness_Score',
```

```
    color_discrete_map={
        'Top 8': '#2E8B57',      # green for happiest
        'Bottom 8': '#C0392B'    # red for least happy
    }
)

# Improve ordering (lowest at bottom, highest at top)
fig.update_layout(
    yaxis=dict(categoryorder='total ascending'),
    xaxis=dict(title='Happiness Score', range=[0, combined['Happiness_Score'].
 □max() + 1]),
    template='plotly_white',
    title='A stark happiness divide: Top countries score nearly twice as high □
 □as the lowest, revealing deep global inequality in well-being '
)

# Add value labels
fig.update_traces(
    texttemplate='%{text:.2f}',
    textposition='outside'
)

# Add vertical line for global average (stretch goal)
fig.add_vline(
    x=global_avg,
    line_dash='dash',
    line_color='black',
    annotation_text=f'Global Avg: {global_avg:.2f}',
    annotation_position='top'
)

# Add a subtle separator annotation between groups
fig.add_annotation(
    x=global_avg,
    y=7.5,  # roughly between bottom 8 and top 8
    text='← Bottom 8 vs Top 8 → ',
    showarrow=False,
    font=dict(size=12, color='gray')
)

fig.show()
```

```
Global average: 5.81
```

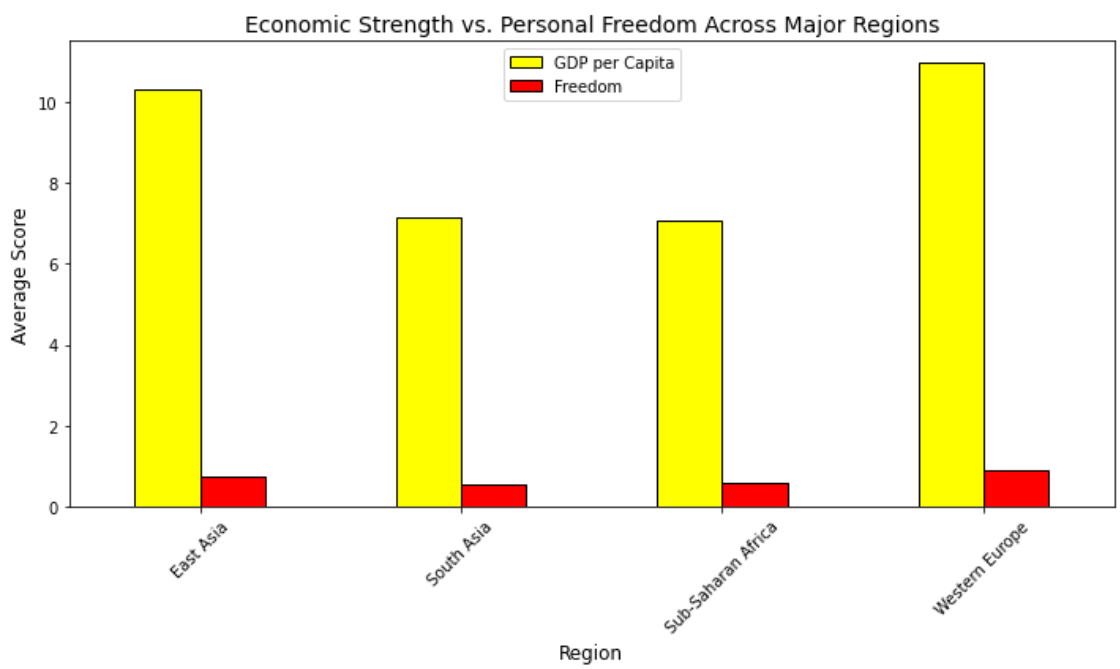


[11]:
```
# Step 1: Filter regions and compute averages for the 2 sub-factors
target_regions = ['Western Europe', 'Latin America', 'East Asia',
'Sub-Saharan Africa', 'South Asia']

# This code automatically finds your GDP and Freedom columns so it
won't crash gdp_col = [c for c in df.columns if 'gdp' in
c.lower()][0] freedom_col = [c for c in df.columns if 'freedom' in
c.lower()][0] columns_to_plot = [gdp_col, freedom_col]

# Filter, group by region, and calculate the mean
stretch_data =
(df[df['Region'].isin(target_regions)]
              .groupby('Region')[columns_to_plot]
              .mean())

print("Data prepared for plotting:")
print(stretch_data)


# Step 2: Build your chart
import matplotlib.pyplot as
plt

# Plotting with yellow and red colors
# kind='bar' creates the grouped bars automatically
stretch_data.plot(kind='bar', figsize=(10, 6), color=['yellow',
'red'], edgecolor='black')

# Adding titles and labels plt.title('Economic Strength vs.
Personal Freedom Across Major Regions',
 fontsize=14)
plt.xlabel('Region', fontsize=12)
plt.ylabel('Average Score',
fontsize=12)
```

```
# Rotating the region names so they are easy to read
plt.xticks(rotation=45)
```

```
# Adding the legend
plt.legend(['GDP per Capita', 'Freedom'])

plt.tight_layout()
plt.show()
```

```
Data prepared for plotting:
                        GDP   Freedom
Region
East Asia      10.312000 0.754000 South
      Asia      7.152250 0.546250
 Sub-Saharan Africa 7.082000 0.599429
 Western Europe    10.963533 0.896067
```



[ ]:

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